DOGECOIN PREDICTION ALGORITHM

Binance-Powered Forecasting Framework

Version 2.0 — DOGE Edition — March 2026

Asset: DOGEUSDT · Exchange: Binance · Primary Interval: 1 Hour

What Changed in v2.0

This framework has been fully revised from the generic crypto template to target DOGECOIN (DOGE)
specifically. Every section reflects DOGE's unique characteristics: its sentiment-driven price action, three distinct historical eras, high BTC beta with periodic decoupling, retail psychology at round numbers, and the critical role of perpetual futures funding rates as a contrarian signal.

New sections: 2 (DOGE Asset Profile) · 3 (Data Extraction Plan) · 5 (Regime System)
Updated: Framework Overview · All feature tables · Model stack · Risk rules · Limitations

1. Framework Overview

The DOGE prediction system is organised into seven interconnected layers. Two layers are new in v2.0: the DOGE Asset Intelligence layer (contextual knowledge) and the Regime Classification layer (market state awareness). Every downstream layer depends on both.

Phase	Name	Key Actions	Output
1	**Asset Intelligence**	Encode DOGE behavioural rules, historical era boundaries, round-number levels, and sentiment proxy definitions	Knowledge base
2	**Data Ingestion**	Extract DOGEUSDT, BTCUSDT, DOGEBTC, 4h/1d, funding rates from Binance REST + WebSocket APIs	Raw multi-source data
3	**Data Processing**	Validate, align all symbols to identical timestamp index, era-partition, store in TimescaleDB	Clean aligned dataset

4	**Regime Classification**	Label every candle with one of 5 sub-regimes; detect transitions; route to regime-specific config	Regime-labelled matrix
5	**Feature Engineering**	Standard TA indicators + 12 mandatory DOGE-specific features (BTC corr, funding, vol spike, HTF, etc.)	Feature matrix
6	**Model Training**	XGBoost-first validation → regime-specific XGBoost → LSTM → Ensemble, all via walk-forward CV	Trained model registry
7	**Signal Generation**	Regime-adjusted threshold inference → risk overrides → logged signal emission	Trade signals

2. DOGE Asset Profile

This section encodes the knowledge the system must have about DOGE before any data is collected or model designed. These are not suggestions — they are structural facts about the asset that directly determine design decisions.

DOGE Identity & Exchange Pairs

Primary pair: DOGEUSDT — Binance Spot (main training and inference target)
Secondary pair: DOGEBTC — Binance Spot (relative strength — altcoin season signal)
Futures pair: DOGEUSDT — Binance USD-M Perpetual Futures (funding rate signal)
Binance listing: ~July 2019
Primary training: January 2022 – Present (post-mania normalisation era)
Context data: July 2019 – December 2021 (stored, not primary training rows)
Primary interval: 1 hour (~26,000 training candles from Jan 2022 to Mar 2026)
All-time high: $0.7310 (May 8, 2021) — used for ath_distance feature

2.1 Three Historical Eras

DOGE's full history contains three statistically distinct eras. Each has different volatility, correlation, and market structure. Mixing eras in training without explicit labelling degrades model quality severely.

DOGE Historical Era Map

Era	Period	Price Range	Key Characteristics	Training Role
DORMANT	Jul 2019 – Dec 2020	$0.002–$0.007	Ultra-low volatility. Tight range. High BTC correlation. Very low retail activity.	Context only. Not in training folds.

MANIA	Jan 2021 – May 2021	$0.007–$0.731	100x parabolic move. Pure sentiment. Fundamentals irrelevant. BTC decoupled.	Excluded entirely from training.
TRANSITION	Jun 2021 – Dec 2021	$0.731–$0.150	Post-peak distribution. High volatility. Capitulation. Unstable regime.	Excluded from training. Too transitional.
CURRENT	Jan 2022 – Present	$0.060–$0.220	Moderate vol. BTC-correlated with periodic decoupling. Consistent structure.	PRIMARY training dataset only.

2.2 Seven Behavioural Facts

These seven facts govern every design decision in the framework. Each maps directly to a mandatory feature, a model rule, or a risk override.

DOGE Behavioural Facts → Design Decisions	
Fact 1 — Sentiment Primacy	Price is more responsive to social media and celebrity mentions than on-chain fundamentals. Pure TA models have systematic blind spots. Mitigation: volume spike detection as highest-available sentiment proxy.
Fact 2 — BTC High Beta	DOGE has rolling BTC beta ~0.8–1.2 in the current era. BTC sells off → DOGE sells harder. BTC rallies → DOGE may outperform or lag. Design: BTC return and BTC volatility are always explicit model inputs.
Fact 3 — Periodic Decoupling	Rolling 24h DOGE-BTC log-return correlation drops below 0.30 during DOGE-specific moves. These decoupling events are the highest-signal moments for DOGE-specific prediction. Design: doge_btc_corr_24h triggers DECOUPLED regime.
Fact 4 — Volume Leads Price	A volume spike (>3× 20-period rolling average) has historically preceded continued directional movement within 4–12 hours. Volume is a leading indicator for DOGE. Design: volume_spike_flag and volume_ratio are mandatory features.
Fact 5 — Funding Rate Extremes	Perpetual funding rate >+0.1%/8h = crowded longs → mean reversion likely. Rate <−0.05%/8h = crowded shorts → short squeeze potential. These are hard override signals, not model inputs to be weighted. Design: funding_extreme_long suppresses ALL BUY signals regardless of model output.
Fact 6 — Altcoin Season Sensitivity	DOGE outperforms strongly when DOGEBTC ratio rises while BTC dominance falls. Design: dogebtc_mom features capture this. When DOGE outperforms BTC in BTC terms, momentum signals weighted higher.
Fact 7 — Round Number Psychology	Retail traders fixate on $0.10, $0.20, $0.50. Price stalls and reverses at these levels more consistently than other assets. Design: distance_to_round_pct and at_round_number_flag reduce position sizing near major levels.

3. Data Extraction Plan — All Binance Sources

All market data is sourced exclusively from Binance. The system requires seven distinct data sources. They must be extracted in the priority order below. BTC data is extracted in parallel with DOGE — not after.

Complete Data Source Inventory

Priority	Source	Binance Endpoint	Interval	Date Range	Est. Rows
1 — CRITICAL	DOGEUSDT Spot 1h	GET /api/v3/klines?symbol=DOGEUSDT&interval=1h	1 hour	Jul 2019 – now	~42,000
2 — CRITICAL	BTCUSDT Spot 1h	GET /api/v3/klines?symbol=BTCUSDT&interval=1h	1 hour	Jul 2019 – now	~42,000
3 — HIGH	DOGEBTC Spot 1h	GET /api/v3/klines?symbol=DOGEBTC&interval=1h	1 hour	Jul 2019 – now	~42,000
4 — MEDIUM	DOGEUSDT Spot 4h	GET /api/v3/klines?symbol=DOGEUSDT&interval=4h	4 hours	Jul 2019 – now	~10,500
5 — MEDIUM	DOGEUSDT Spot 1d	GET /api/v3/klines?symbol=DOGEUSDT&interval=1d	1 day	Jul 2019 – now	~1,750
6 — HIGH	DOGE Funding Rate	GET /fapi/v1/fundingRate?symbol=DOGEUSDT&limit=1000	8 hours	Oct 2020 – now	~5,400
7 — LOW (live)	DOGEUSDT Agg Trades	GET /api/v3/aggTrades?symbol=DOGEUSDT	Tick	Last 30 days only	~5M+

3.1 REST API — Rate Limits & Best Practices

REST API Rules

Base URL (Spot)	https://api.binance.com
Base URL (Futures)	https://fapi.binance.com — for funding rate endpoint only

Weight Limit	1,200 request weight per minute. Monitor X-MBX-USED-WEIGHT-1M header on every response.
Klines Weight	Each /api/v3/klines call costs 2 weight. Max 1,000 candles per call — paginate for longer ranges.
Funding Rate Weight	Each /fapi/v1/fundingRate call costs 1 weight. Max 1,000 records per call.
Rate Limit Safety	Pause 100ms between requests. If weight > 960 (80%), sleep until next minute boundary.
Retry Policy	Retry 5xx errors: 5 attempts with 2^n backoff (1s,2s,4s,8s,16s). Retry 429 after Retry-After duration. Never retry 4xx.
API Key Scope	Use a read-only API key scoped to market data. Store in secrets.env. Never log, print, or commit.
Checkpoint Rule	Save bootstrap progress checkpoint every 5,000 rows per symbol. Resume from checkpoint on failure.
Timestamp Format	All timestamps: UTC epoch milliseconds. Use close_time (not open_time) as the reference for a completed candle.

3.2 WebSocket API — Live Streaming

WebSocket Streams — DOGE Configuration	
DOGE Kline Stream	wss://stream.binance.com:9443/ws/dogeusdt@kline_1h — primary live candle stream
BTC Kline Stream	wss://stream.binance.com:9443/ws/btcusdt@kline_1h — required for BTC correlation features
DOGE Book Ticker	wss://stream.binance.com:9443/ws/dogeusdt@bookTicker — best bid/ask for spread feature
DOGE Agg Trades	wss://stream.binance.com:9443/ws/dogeusdt@aggTrade — for live CVD approximation
Candle Filter	Only process messages where kline.x == true (candle closed). Ignore all partial candle updates.
Reconnection	Auto-reconnect with exponential backoff. On reconnect: REST-fetch missed candles before resuming inference.
Watchdog	Thread checks last_message_time every 30s for ALL streams. If any stream stale > 2× interval: alert + suspend inference.
Gap Recovery	On reconnect: fetch REST klines for gap period. Validate gap-fill. Resume inference only after buffer refilled.
Multi-Stream Note	Both DOGE and BTC streams must be healthy before any inference runs. BTC stream failure suspends DOGE inference.

3.3 Multi-Symbol Alignment

After bootstrap, all three primary symbols (DOGEUSDT, BTCUSDT, DOGEBTC) must share an identical timestamp index. This is enforced by the aligner module before any feature engineering begins.

⚠ Alignment Rules & Edge Cases

- Inner-join all three DataFrames on open_time first. Investigate any symbol with fewer rows than the intersection.
- DOGEBTC occasionally has maintenance gaps absent from DOGEUSDT. Forward-fill DOGEBTC price ONLY (not volume)

  for gaps <= 3 consecutive candles. Tag as dogebtc_interpolated=True.
- For DOGEBTC gaps > 3 candles: do not forward-fill. Set DOGEBTC-derived features to NaN for that period.

  Mask those NaN features before model inference — do not impute with zero.
- The funding rate (8h) must be forward-filled to the 1h OHLCV index:

  each 8h funding rate applies to the 8 hourly candles it covers until the next rate is published.
- BTC data is used as a feature source only — never as a training target or as part of any DOGE prediction.

4. Data Processing & Storage

4.1 Validation Contract

Every data batch — from bootstrap or live WebSocket — must pass all validation checks before being written to storage. Validation failures halt the pipeline entirely.

Mandatory Validation Checks	
Schema	Columns: open_time, open, high, low, close, volume, close_time, quote_volume, num_trades. Reject batches with missing or extra columns.
Timestamp Monotonic	df['open_time'].is_monotonic_increasing must be True. Reject entire batch if False.
Interval Consistency	Consecutive open_time gap must equal interval_ms exactly. Gap > 3 candles triggers pipeline halt.
OHLCV Sanity	high >= open, high >= close, low <= open, low <= close, close > 0, volume >= 0. Tag violations — never silently drop.
No NaN / Inf	After cleaning: zero NaN or Inf in any OHLCV column. Hard requirement — not a warning.
Multi-Symbol Alignment	Post-aligner: DOGEUSDT, BTCUSDT, DOGEBTC must have identical timestamp index. Assert before features run.

Funding Rate Interval	Consecutive funding rate timestamps must be exactly 28,800,000ms (8h) apart. Any other gap is a data error.
Era Partitioning	Pre-2022 rows tagged era='context'. Post-2022 rows tagged era='training'. Never mixed in same Parquet partition.
UTC Enforcement	All timestamps as UTC epoch ms. Reject tz-naive datetime objects at validator boundary.
Stale Data Check	At inference time: last candle close_time must be within 2 × interval_ms of now. Halt if stale.

4.2 Storage Architecture

Storage Layer Design	
DOGEUSDT OHLCV	TimescaleDB hypertable: dogeusdt_1h. Partitioned by month. Index on open_time. Append-only after bootstrap.
BTCUSDT OHLCV	TimescaleDB hypertable: btcusdt_1h. Separate table — never merged with DOGE OHLCV columns.
DOGEBTC OHLCV	TimescaleDB hypertable: dogebtc_1h. Stores raw DOGEBTC pair price for relative strength computation.
Funding Rates	TimescaleDB hypertable: doge_funding_rates. Columns: timestamp_ms, funding_rate, mark_price, funding_available_flag.
Feature Store	Apache Parquet — partitioned by era (context/training) and regime label. Versioned by run_id. Schema validated via pandera.
Regime Labels	TimescaleDB table + Parquet: timestamp_ms, regime_label, btc_corr_24h, bb_width, atr_norm, regime_transition_flag.
Model Registry	MLflow artifact store: model weights + scaler.pkl + feature_columns.json + regime_config.yaml per run.
Signal Log	Append-only TimescaleDB table: timestamp, regime, lstm_prob, xgb_prob, ensemble_prob, threshold_used, signal, confidence.
Live Feature Buffer	In-memory ring buffer (deque, maxlen=500) of last 500 closed 1h candles for real-time inference.

5. DOGE Market Regime Classification

Regime classification is mandatory. It must run before any feature engineering and before every live inference. A model that does not know its current regime will apply the wrong confidence threshold and wrong position sizing.

Five DOGE Sub-Regimes (Post-2022 Current Era)

TRENDING_BULL — EMA20 > EMA50 > EMA200. Price above all. ATR_norm > 0.4%. 7d return > +5%.

Model: momentum features weighted higher. Mean-reversion signals suppressed.

Confidence threshold: 0.62

TRENDING_BEAR — EMA20 < EMA50 < EMA200. Price below all. ATR_norm > 0.4%. 7d return < -5%.

Model: momentum features weighted higher (bearish). RSI oversold signals ignored.

Confidence threshold: 0.62

RANGING_HIGH_VOL — Price within Bollinger Bands. BB width > 0.04. ATR_norm > 0.5%. 7d return -5% to +5%.

Model: mean-reversion signals active. Momentum signals suppressed.

Confidence threshold: 0.65

RANGING_LOW_VOL — BB width < 0.04. ATR_norm < 0.3%. 7d return near zero.

Model: low-confidence regime. Reduce position sizing by 50%. Increase threshold.

Confidence threshold: 0.70

DECOUPLED — Rolling 24h DOGE-BTC log-return correlation < 0.30. DOGE-specific move active.

OVERRIDES all other regimes when active. Maximum alert state.

Model: suppress BTC-correlated features. Amplify DOGE-specific features.

Confidence threshold: 0.72 | Position sizing: 50% of base

Regime Classification Rules	
Priority Order	DECOUPLED is checked first and overrides all other regime labels when active.
Compute Frequency	Re-run classifier on every closed 1h candle. Regime label is dynamic, not static.
Transition Logging	Every regime change must be logged: from_regime, to_regime, trigger_features, timestamp.
Stabilisation Buffer	After transitioning FROM DECOUPLED: wait 3 candles before resuming normal threshold (stabilisation period).
Regime Features	regime_label is one-hot encoded and included as explicit model input to LSTM and ensemble meta-learner.
Per-Regime Models	Train a separate XGBoost model per sub-regime. LSTM trained on all regimes combined (learns transitions).
Regime Sampling	Within each walk-forward training fold: if any sub-regime has < 5% representation, oversample to 5% minimum.

Regime Archiving	Save regime distribution (% per label) per walk-forward fold to MLflow. Enables auditing of training balance.

6. Feature Engineering

Features are built in two layers. The standard layer applies to any crypto asset. The DOGE-specific layer contains 12 mandatory features that encode DOGE's unique behavioural characteristics. Both layers are required.

6.1 Standard Technical Features

Price-Based Indicators	
SMA / EMA	Periods: 7, 14, 21, 50, 200 on close. Also: close/EMA200 ratio (distance from long-term trend).
MACD	12/26/9. Features: macd_line, signal_line, histogram, histogram_direction (sign-change flag).
RSI	Period 14. Features: rsi_value, rsi_overbought (>70 flag), rsi_oversold (<30 flag), rsi_divergence.
Bollinger Bands	20-period, 2 std. Features: bb_pct_b, bb_width, bb_squeeze_flag (width < 0.04).
ATR	Period 14. Store raw ATR and ATR_norm (ATR/close). Use normalised for all cross-period comparisons.
Stochastic	14/3/3. Features: stoch_k, stoch_d, stoch_crossover_flag.
Ichimoku	Tenkan(9), Kijun(26), Senkou A/B. Feature: cloud_position (+1 above / -1 below / 0 inside cloud).

Volume & Lag Features	
OBV	On-Balance Volume. Also: obv_ema_ratio (OBV / 20-period EMA of OBV).
VWAP	Resets at UTC midnight daily. Feature: price_vs_vwap = (close - VWAP) / VWAP.
CMF	Chaikin Money Flow, 20-period.
CVD (approx)	Historical: (close-low)/(high-low) × volume cumulative sum. Live: from aggTrade stream.
Log Returns	log(close[t]/close[t-n]) for n = 1, 3, 6, 12, 24, 48, 168. Primary price-change inputs.

Rolling Volatility	std(log_returns) over windows: 6, 12, 24, 48, 168. Normalised by rolling mean.
Momentum	close[t]/close[t-n] - 1 for n = 6, 12, 24, 48. Trend strength signal.
High-Low Range	(high - low) / close. Normalised intrabar range — indecision proxy.
Rolling Skewness / Kurtosis	24-period window. Kurtosis spike often precedes fat-tail DOGE moves.

6.2 DOGE-Specific Mandatory Features

All 12 of these features must be present in every training run. A training run missing any of them is considered incomplete and must not be promoted to production.

12 Mandatory DOGE-Specific Features

Feature Name(s)	Source Module	Formula / Logic	What It Captures
doge_btc_corr_12h, _24h, _7d	doge_specific.py	doge_log_ret.rolling(N).corr(btc_log_ret) for N=12,24,168	BTC-DOGE coupling strength. <0.30 triggers DECOUPLED regime.
dogebtc_mom_6h, _24h, _48h	doge_specific.py	log(dogebtc_close[t] / dogebtc_close[t-N]) for N=6,24,48	DOGE outperformance vs BTC in BTC terms — altcoin season proxy.
volume_ratio	doge_specific.py	volume / volume.rolling(20).mean()	Normalised volume surge. Non-stationary raw volume is banned.
volume_spike_flag	doge_specific.py	(volume_ratio >= 3.0).astype(int)	Binary leading indicator. Spike precedes directional move 4-12h.
volume_spike_magnitude	doge_specific.py	volume_ratio.clip(upper=10.0) / 10.0	Continuous spike intensity [0,1] — complements binary flag.
funding_rate	funding_features.py	Raw 8h perpetual funding rate, forward-filled to 1h index	Absolute leverage

			positioning level.
funding_rate_zscore	funding_features.py	(fr - fr.rolling(90).mean()) / fr.rolling(90).std()	Relative funding extreme — comparable across time periods.
funding_extreme_long	funding_features.py	(funding_rate > 0.001).astype(int)	Crowded long flag. HARD OVERRIDES all BUY signals when active.
funding_extreme_short	funding_features.py	(funding_rate < -0.0005).astype(int)	Crowded short flag. Short squeeze potential signal.
htf_4h_rsi, htf_4h_trend, htf_4h_bb_pctb	htf_features.py	4h indicators, forward-filled to 1h — use LAST CLOSED 4h candle	Macro momentum context above 1h noise.
htf_1d_trend, htf_1d_return, ath_distance	htf_features.py	1d EMA trend, 1d return, log(ATH/$0.731 / close)	Daily directional context and distance from all-time peak.
distance_to_round_pct, at_round_number_flag, nearest_round_level	doge_specific.py	Round levels: [0.05,0.10,0.15,0.20,0.25,0.30,0.50,1.00]	Retail psychological S/R levels. Reduces position sizing near levels.

⚠ Lookahead Bias — The Most Dangerous Feature Error

Lookahead bias occurs when a feature at time T uses information from time T+1 or later. It inflates backtest accuracy to 80%+ — values that are completely unreproducible live.

Three DOGE-specific lookahead traps:

- HTF candle not yet closed: at 15:00, the current 4h candle closes at 16:00. Use the 4h candle that closed at 12:00. Forward-fill until the 16:00 boundary, then update.
- BTC correlation on prices, not returns: raw price correlation is non-stationary and spurious. Always compute on log returns. Write a unit test asserting this.
- Volume spike on raw volume: DOGE volume trends upward over years. Raw volume in 2019 is incomparable to 2024 volume. Always normalise: volume / rolling(20).mean().

7. Model Architecture

7.1 Model Build Order

The build order is mandatory. Models are added incrementally — each one only after the previous has been validated. This prevents wasting time on complex models when the feature layer is the actual problem.

✓DOGE Model Build Sequence

STEP 1 — Global XGBoost on all post-2022 data (single model).
GATE: must achieve directional accuracy > 53% out-of-sample.
If it fails this gate, STOP. The features do not contain sufficient signal.
Fix the feature layer before proceeding. Do not skip to LSTM.

STEP 2 — Regime-specific XGBoost (one model per sub-regime).
Compare each against global XGBoost on its own regime's test data.
Promote regime-specific model only if it outperforms global on that regime.

STEP 3 — LSTM on full post-2022 data, with regime_label_encoded as explicit input.
LSTM learns temporal patterns across regime transitions that XGBoost cannot capture.

STEP 4 — Transformer: build ONLY if LSTM Sharpe Ratio < 1.0. Otherwise skip.
The Transformer adds engineering complexity for marginal DOGE-specific benefit.

STEP 5 — Ensemble meta-learner: inputs = [xgb_regime_prob, lstm_prob, regime_label_encoded].
Use logistic regression or 2-layer MLP. Train on walk-forward validation fold outputs.

7.2 Model Specifications

LSTM Architecture	
Input Shape	(batch, sequence_length=60, n_features) — 60-candle 1h lookback window
Layer 1	LSTM(128 units, return_sequences=True, dropout=0.2, recurrent_dropout=0.1)
Layer 2	LSTM(64 units, return_sequences=False, dropout=0.2)
Layer 3	Dense(32, relu) + BatchNorm + Dropout(0.3)
Output	Dense(1, sigmoid) — binary probability of upward direction

Loss	BCELoss (classification). MSELoss only for regression target variant.
Optimiser	Adam(lr=1e-3, weight_decay=1e-5) + ReduceLROnPlateau(patience=5, factor=0.5)
Early Stopping	Monitor val_loss. Patience=15. Restore best weights.
Gradient Clip	clip_grad_norm_(model.parameters(), max_norm=1.0) — prevents NaN loss on DOGE vol spikes
Batch Size	64 (reduce to 32 if memory constrained). Never use batch_size=1.

XGBoost Parameters	
Objective	binary:logistic
n_estimators	500 with early_stopping_rounds=20 on validation set
learning_rate	0.05 — deliberately low
max_depth	5 — conservative for crypto noise
subsample / colsample_bytree	0.8 / 0.8
scale_pos_weight	count(neg) / count(pos) — recalculated each fold
tree_method	hist (CPU) or gpu_hist (CUDA)
SHAP Analysis	Run SHAP value analysis after training. Log top-20 features to MLflow. Verify DOGE-specific features in top-10.

7.3 Training Configuration

DOGE Walk-Forward Training Parameters

```
Primary dataset:        DOGEUSDT 1h  |  January 1 2022 – present  |  ~26,000 candles
training_window_days:   180   (6 months = ~4,320 candles per fold)
validation_window_days: 30    (1 month  = ~720 candles per fold)
step_size_days:         7     (advance 1 week per fold)
min_training_rows:      3,000 (minimum before first fold begins)

Era enforcement:        Pre-2022 rows must NOT appear in any training fold.
                        Assert: min(X_train.index) >= '2022-01-01' before every fit.

Regime stratification:  Within each fold, compute regime distribution.
                        If any sub-regime < 5% of fold: oversample that regime to 5%.
                        Log regime distribution per fold to MLflow.

Scaler rule:            Fit StandardScaler on X_train ONLY. Transform X_val and X_test
```

```
                    using training fold scaler. Re-fit scaler at every walk-forward step.
                    Save scaler.pkl with model in MLflow artifact store.

Seed:               random=42, numpy=42, torch=42. Set before EVERY training run.
                    Verify reproducibility: two identical runs produce identical weights.
```

7.4 Ensemble & Regime-Adjusted Signal Threshold

✓Signal Generation Logic

```
1. Classify current regime using DogeRegimeClassifier.
2. Load confidence_threshold from regime_config.yaml using current regime key:
   TRENDING_BULL / TRENDING_BEAR:  0.62
   RANGING_HIGH_VOL:               0.65
   RANGING_LOW_VOL:                0.70
   DECOUPLED:                      0.72

3. Run ensemble:
   ensemble_prob = meta_learner.predict_proba([xgb_regime_prob, lstm_prob, regime_encoded])

4. Apply risk overrides BEFORE emitting signal (see Section 9):
   if funding_extreme_long == 1:  SUPPRESS BUY regardless of ensemble_prob
   if at_round_number_flag == 1:  reduce position size by 30% on BUY signals
   if regime == RANGING_LOW_VOL:  reduce position size by 50%
   if regime == DECOUPLED:        reduce position size by 50%

5. Apply threshold:
   ensemble_prob >= threshold   → emit BUY
   ensemble_prob <= 1-threshold → emit SELL (if shorts enabled in config)
   otherwise                    → emit HOLD

6. HOLD is a valid, intentional output. Never force a direction.
```

8. Validation & Backtesting

8.1 Walk-Forward Validation

Standard k-fold cross-validation is completely invalid for time-series prediction. Walk-forward is the only acceptable methodology. This rule has no exceptions.

- Training folds are strictly in the past relative to validation folds — no temporal overlap ever
- Scaler is re-fitted on each training fold. It never sees validation or test data before fitting
- Hyperparameter tuning (Optuna) runs on training+validation folds only. Test set untouched until final evaluation
- Test set is examined exactly once after all optimisation is complete. Looking at it during tuning invalidates it

8.2 Realistic Execution Model

Backtesting Execution Requirements	
Fill Price	Entry and exit fill at NEXT candle's open. Never fill at the signal candle's close — that price is unavailable when the signal fires.
Taker Fee	0.10% per trade per leg. Both entry and exit. Deduct from PnL before any metric computation.
Slippage	Uniform random slippage: 0.02%–0.08% per trade side.
Position Sizing	Fixed fractional: 1% of current equity per trade. Recalculate for every trade. ATR-adjust in RANGING_LOW_VOL (0.5% risk).
Round Number Sizing	Reduce position to 0.7% of equity on BUY signals where at_round_number_flag == 1.
Max Drawdown Halt	Halt simulation (not just log) if drawdown exceeds 25%. Prevents silent catastrophic run.
Per-Regime Reports	Mandatory: run separate performance report per sub-regime. Overall metrics alone are insufficient.
Buy-and-Hold Baseline	Always compare to passive DOGE buy-and-hold over the same test period. Model must outperform on risk-adjusted basis (Calmar).
BTC Drawdown Period	If test period contains BTC drawdown > 20%: report model performance DURING that period separately.

8.3 Performance Acceptance Gates

DOGE Model Acceptance Gates			
Gate	**Metric**	**Threshold**	**DOGE Note**
G1	Directional Accuracy (OOS)	>= 54%	Measured on post-2022 test set only. Pre-2022 excluded.
G2	Sharpe Ratio (annualised)	>= 1.0	Must also be >= 0.8 per regime. No regime can be catastrophic.

G3	Max Drawdown	<= 20%	DOGE amplifies BTC drawdowns. 20% is tight but achievable.
G4	Calmar Ratio	>= 0.6	Higher than generic threshold due to DOGE volatility profile.
G5	Profit Factor	>= 1.3	Gross profit / gross loss across all trades.
G6	Win Rate	>= 45%	Acceptable if avg_win substantially exceeds avg_loss.
G7	Trade Count (test period)	>= 150	More trades needed for statistical validity given DOGE noise.
G8	vs Buy-and-Hold	Calmar > BH Calmar	Must beat passive DOGE holding on risk-adjusted basis.
G9	DECOUPLED Regime Drawdown	<= 15%	Decoupled events are high-risk. Reduced sizing must contain losses.

9. Risk Management & Safeguards

The prediction framework outputs probabilities, not guarantees. All signals must pass through this risk layer before reaching any execution consumer. Risk rules are not suggestions — they are hard overrides.

Core Risk Rules	
Max Position Risk	Never risk more than 1% of current equity per trade. 0.5% in RANGING_LOW_VOL. 0.7% near round numbers.
Daily Loss Limit	Halt all signal emission if intraday drawdown exceeds 5% of starting equity. Resume next calendar day only.
Funding Override	If funding_extreme_long == 1: SUPPRESS ALL BUY SIGNALS. Model probability is irrelevant. Hard rule.
Round Number Reduction	If at_round_number_flag == 1: reduce position size by 30% on BUY signals. Do not block signal.
Decoupled Sizing	If regime == DECOUPLED: reduce position size to 50% of base. Increase threshold to 0.72.
Signal Cooldown	Minimum 3 candles (3 hours) between new entry signals. No stacking of signals.
ATR Spike Filter	If ATR_norm > 3× its 168-period rolling mean: suppress ALL signals. Extreme volatility environment.
BTC Crash Filter	If BTC 1h log return < -4%: suppress all DOGE BUY signals for 2 candles. BTC crash amplifies to DOGE.
Drawdown Stop	If live equity drawdown exceeds 10% from peak: switch to HOLD mode. Human review before resuming.

Human Override	Dashboard kill switch must be present and tested. Immediately pauses all signal emission.

10. Live Deployment Architecture

10.1 System Components

Deployment Components	
Multi-Stream Ingestion	Asyncio process subscribed to DOGE + BTC WebSocket streams. Both must be healthy before inference runs.
Feature Pipeline	Triggered on every closed 1h candle. Reads last 500 candles from live buffer. Computes all 12 DOGE features.
Regime Classifier	Runs on every candle after feature pipeline. Outputs current regime label. Logs any regime transitions.
HTF Updater	Checks 4h and 1d candle boundaries on every 1h candle. Updates HTF features when new HTF candle closes.
Inference Engine	Loads regime-appropriate XGBoost + LSTM from MLflow. Applies regime threshold. Runs risk overrides.
Signal Router	Receives signals. Applies position sizing rules. Routes to alert consumer (Telegram / webhook).
Monitoring Dashboard	Grafana: regime label, doge_btc_corr_24h, volume_ratio, funding_rate_zscore, inference latency, equity curve.
Alerting	DECOUPLED alert (critical), drawdown halt, BTC crash filter active, accuracy degradation, stream health alerts.

10.2 Latency Budget

- WebSocket tick receipt to feature computation: < 150ms (multi-symbol adds overhead vs single-symbol)
- Feature computation to regime classification: < 30ms
- Regime classification to inference: < 50ms
- Inference to signal emission: < 20ms
- End-to-end (closed candle to signal): target < 300ms for 1h bars

10.3 Retraining Strategy

Retraining Rules	
Scheduled Retraining	Every Sunday 00:00 UTC. Retrain on last 180 days of post-2022 data. Shadow 48h before promoting.
Accuracy Trigger	If rolling 48h directional accuracy < 50%: trigger immediate retraining investigation. Switch to HOLD.
Regime Shift Trigger	Regime transitions alone do NOT trigger retraining. Only accuracy degradation does.
Shadow Mode	New model runs in shadow (predict, don't execute) for minimum 48h before promotion.
Promotion Gate	New model Sharpe > current production model Sharpe on last 30 days. All Section 8.3 gates must pass.
Rollback	Previous model tagged 'previous-production' in MLflow. Rollback completes in < 5 minutes. Test this.
Pre-2022 Rule	Pre-2022 era rows must never appear in any retraining fold. Assert era filter on every retrain.

11. Technology Stack

Recommended Stack — DOGE Edition	
Language	Python 3.11 or 3.12. No other language. The entire ML ecosystem is Python-first.
Data Ingestion	python-binance 1.0.19 (spot REST + WebSocket). Separate Futures client for funding rates via fapi endpoints.
Data Storage	TimescaleDB (OHLCV + funding rates + signals). Apache Parquet (features). MLflow (models + scalers).
ML Framework	PyTorch 2.3.x (LSTM + Transformer). XGBoost 2.0.x. scikit-learn 1.4.x (meta-learner, scalers).
Feature Engineering	ta-lib 0.4.x (price indicators — requires C library). pandas-ta 0.3.x (pure-Python fallback + funding indicators).
Schema Validation	pandera 0.19.x — enforced on every DataFrame read and write. Schema hash stored per Parquet run_id.
Experiment Tracking	MLflow 2.13.x — all training runs, parameters, metrics, model weights, scalers, feature lists.
Hyperparameter Search	Optuna 3.6.x — Bayesian optimisation on train+val folds only.
Orchestration	APScheduler (initial deployment). Migrate to Apache Airflow after pipeline is stable.

Monitoring	Prometheus + Grafana. Custom gauges: regime, doge_btc_corr, funding_zscore, volume_ratio, equity_drawdown.
Backtesting	Custom walk-forward engine (VectorBT as reference). Must support per-regime reporting natively.
Deployment	Docker + Docker Compose. One container per service: ingestion, features+inference, monitoring.
Statistical Tests	statsmodels (ADF stationarity test). scipy (Durbin-Watson, Bonferroni). Run before every training job.

12. Limitations & Important Caveats

⚠ General Limitations

- Crypto markets are non-stationary: patterns learned in one regime may not transfer to the next.
- Black swan events — exchange hacks, regulatory bans, whale manipulation, celebrity announcements — cannot be modelled from price and volume data alone. No technical model predicts these.
- Overfitting risk is severe. Always validate on truly out-of-sample data. Never examine the test set more than once. Every extra look at the test set makes its metrics meaningless.
- This framework produces probabilistic signals — not certainties. No algorithm is profitable 100%.
- Past backtest performance does not guarantee future live results. Regime shifts will occur.
- This document does not constitute financial advice. Always test with paper trading first.

DOGE-Specific Limitations

- The 2021 mania (100x move) will not recur in its original form. Any model trained on that era will have learned a pattern that is not repeatable at the same magnitude. This is why the primary training window is restricted to January 2022 onwards.

- DOGE's price is uniquely sensitive to external sentiment events that have no signature in OHLCV data before they occur. A Elon Musk tweet can move DOGE 20% in minutes — the model will always be reactive to such events, never predictive. Volume spikes are the only available leading indicator for sentiment events.

- The funding rate signal requires futures market participation. During periods of low futures interest, funding rates are near zero and carry no signal. The funding_available_flag must be tracked and features masked during extended near-zero funding periods.

- DOGE's round number psychology is a real but soft phenomenon — it increases probability of rejection at those levels, it does not guarantee it. Position size reduction near round

numbers is risk management, not a trading signal.

- The DECOUPLED regime is the most unpredictable and highest-risk state. When DOGE is moving independently of BTC, the move is almost always sentiment-driven and cannot be reliably predicted by technical features alone. The correct response is caution, not aggression.